Faculty of Engineering, Department of Computer Science

UCL

ENGF0034: Design and Professional Skills

Introductory lecture



Prof. Martin Benning
6 October 2026

ENGF0034 2026/27 is a team effort



Charlotte Maughan Jones



James Cole



Vivek Ramachandran

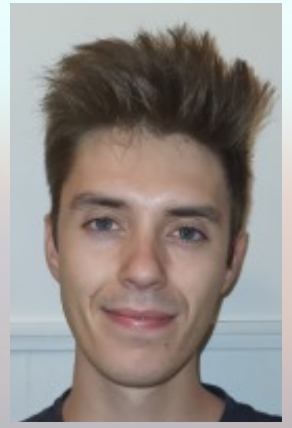


Amy Lourenço



Sunny Bains

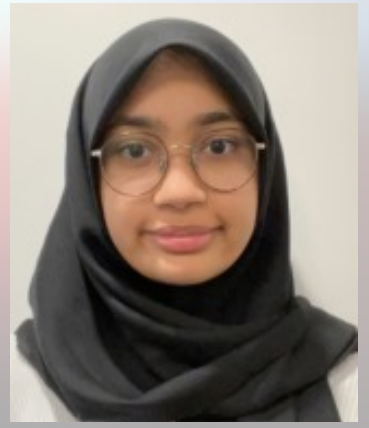
Teaching Assistants



Cris Tudosie



Anya Lakhani

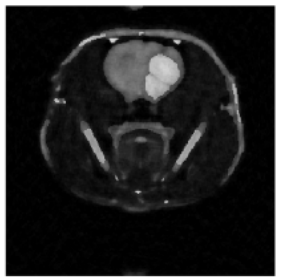


Shafiyyah Shafiyyah

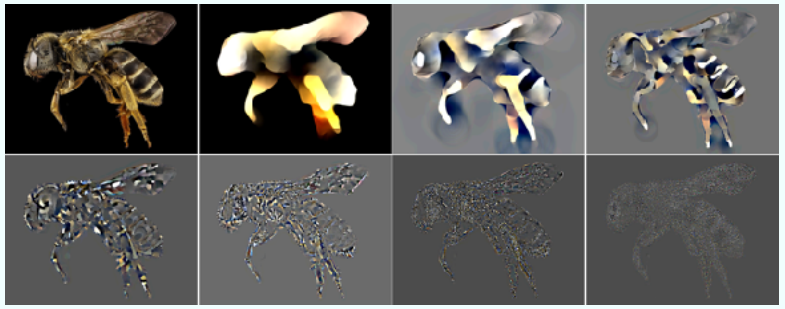
Who am I? Professor of Inverse Problems



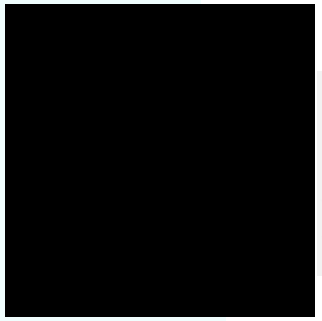
Martin Benning



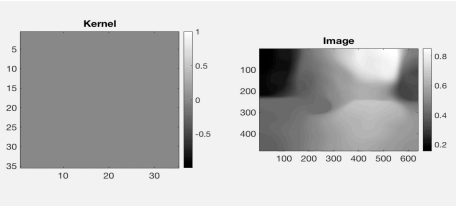
Joint reconstruction & segmentation



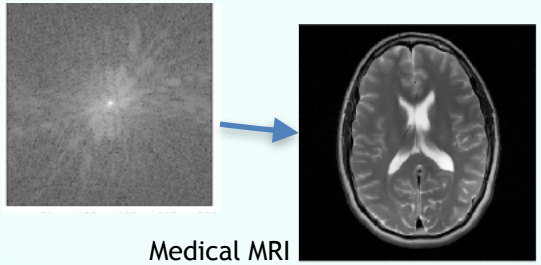
Nonlinear spectral decomposition



Blind image deconvolution

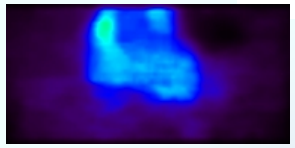




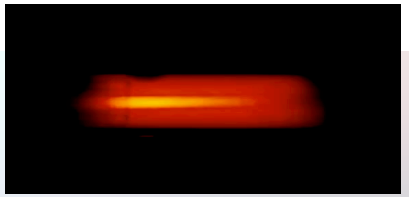


Medical MRI

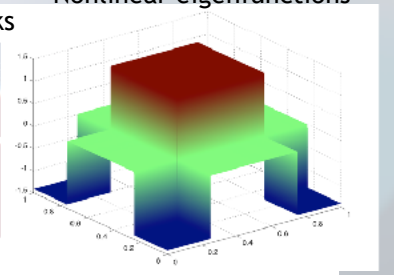
Medical PET



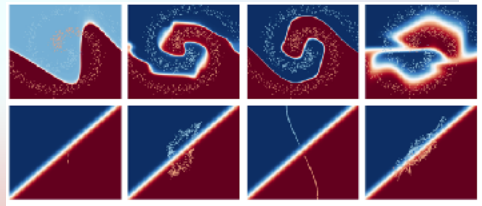
Industrial MRI



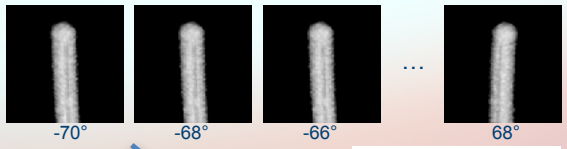
Nonlinear eigenfunctions



Classification with residual neural networks



Transmission Electron Microscopy / Tomography



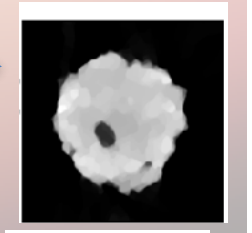
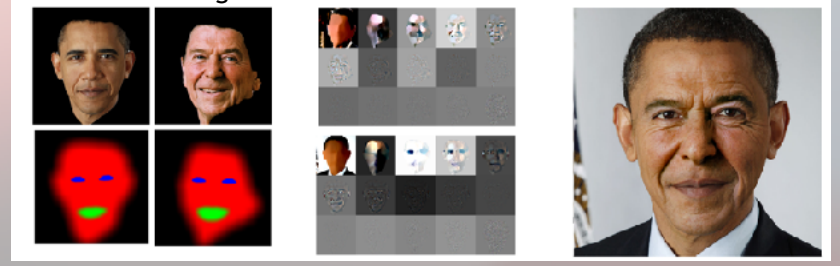


Image reconstruction

Nonlinear image fusion

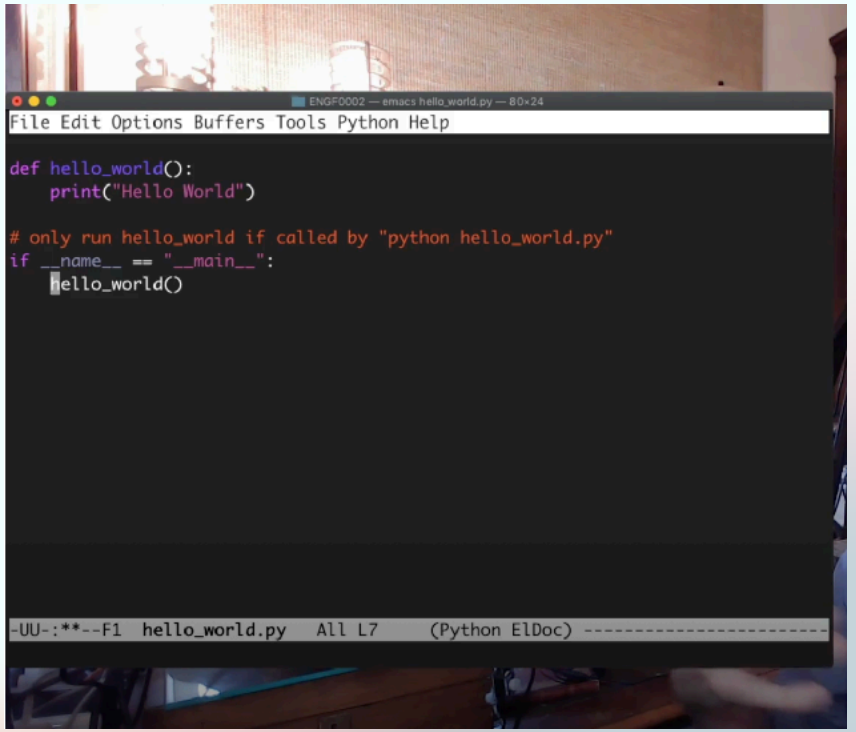


Areas of expertise:

- Inverse & Ill-posed problems
- Regularisation theory
- Machine learning
- Optimisation
- Industrial & medical Imaging
- Image processing

Module approach

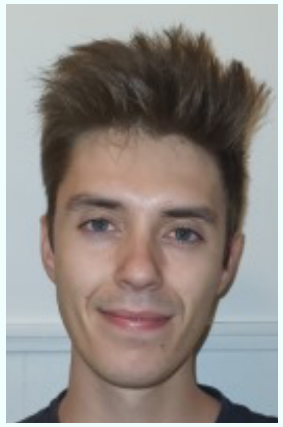
- Lectures (almost) every Tuesday
- Tutorials (almost) every Friday
- Ask and answer questions via the Moodle forum
- Course materials will be released via git
- Assessments will be released via git or Moodle
- Drop-in clinics will take place as extra support for those new to programming

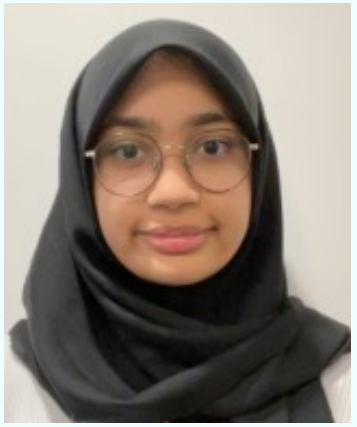


Let us checkout the module Moodle & git pages

Drop-in clinics

Teaching Assistants


Cris Tudosie


Shafiyyah Shafiyyah


Anya Lakhani

- Drop-in clinic Cris (in-person): Wednesday, 3-4 pm
  G03, Henry Morley
- Drop-in clinic Shafiyyah (in-person): Thursday, 4-5 pm
  G03, Henry Morley
- Drop-in clinic Anya (in-person): Friday 1-2 pm
  G03, Henry Morley

(Starts next week!)

Assignments

- Week 1 - 2: Introductory programming assignment
- Week 2 - 3: Debug the bomber game
- Week 3 - 5: Debug the frogger game
- Week 5 - 7: Write a tetris autoplayer (20%)
- Week 7 - 9: Write pacman protocol specification (15%)
- Week 7 in Term 2: Ethics report (15%)
- Term 2 Reading Week: Presentations (10%)
- + Scenario assessments in Term 2 (remaining 40%)

Assignments

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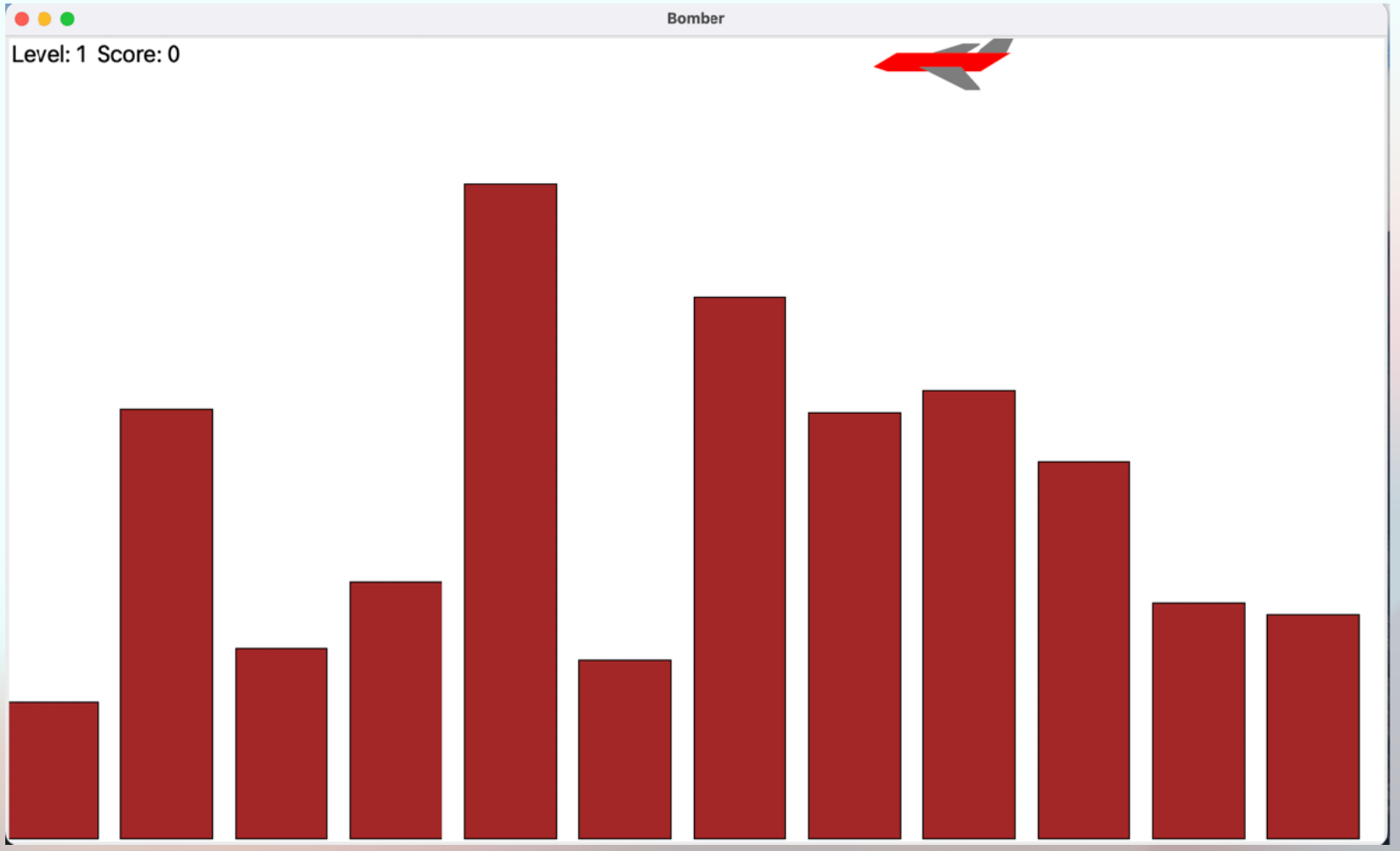
Do not plan time away during reading week!

Assignments

Assignment 2: (Canyon) Bomber

Classic Atari Arcade Game from 1977

Task: Find five bugs that make the game unplayable, and fix them!

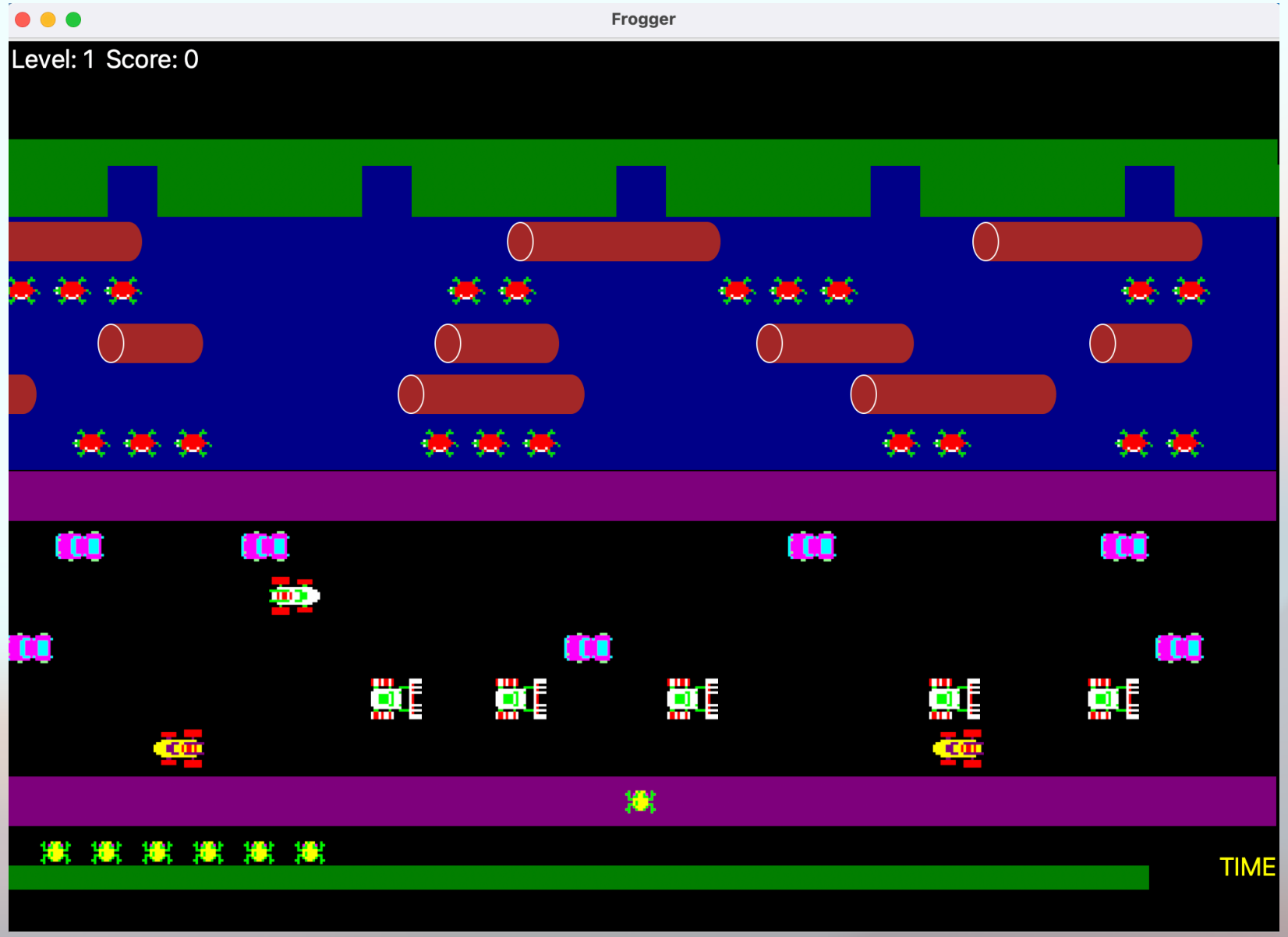


Assignments

Assignment 3: Frogger

Classic Konami/Sega Arcade Game from 1981

Task: Find five bugs that make the game unplayable, and fix them!



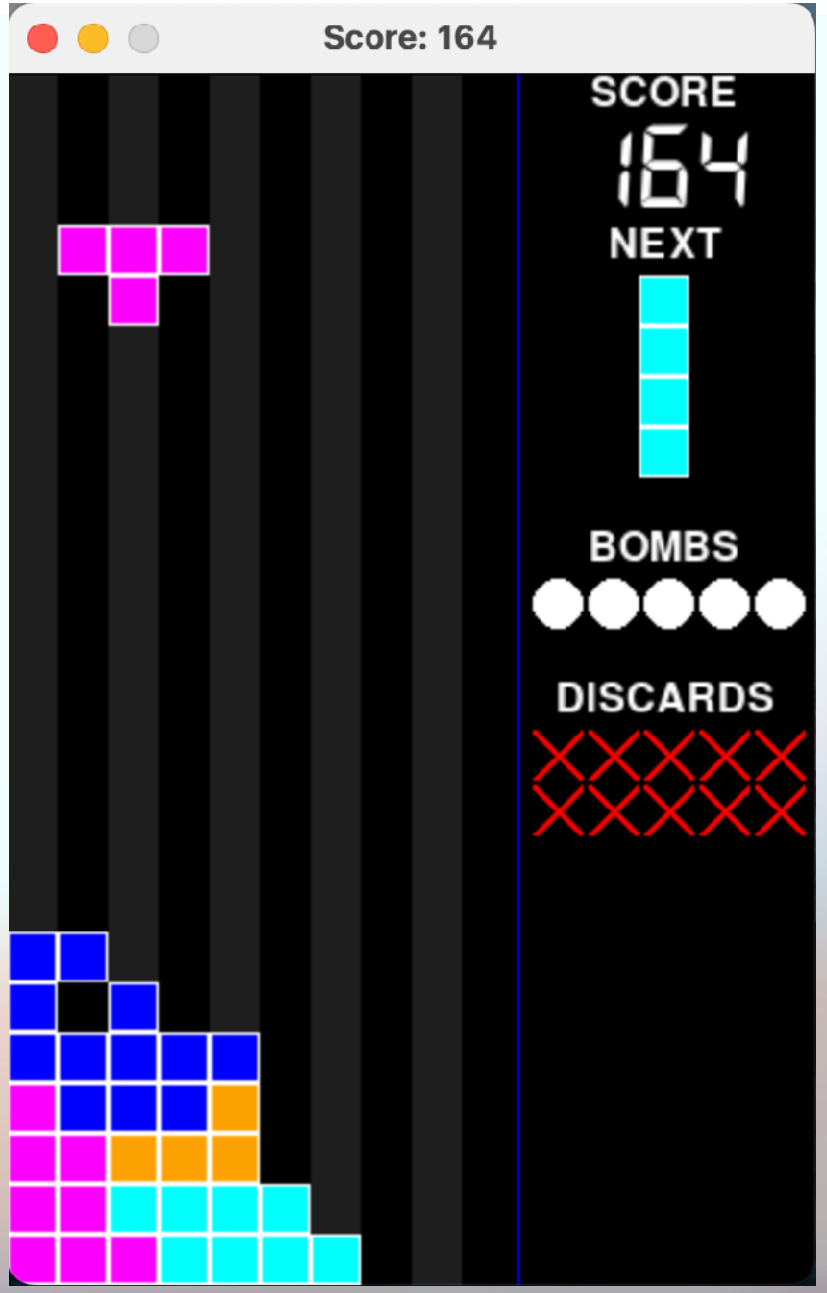
Assignments

Assignment 4: Tetris autoplayer

Puzzle video game created by Alexey Pajitnov in 1985

Game Boy version by Nintendo popularised Tetris in 1989

Task: write a program that plays Tetris and compete with your peers



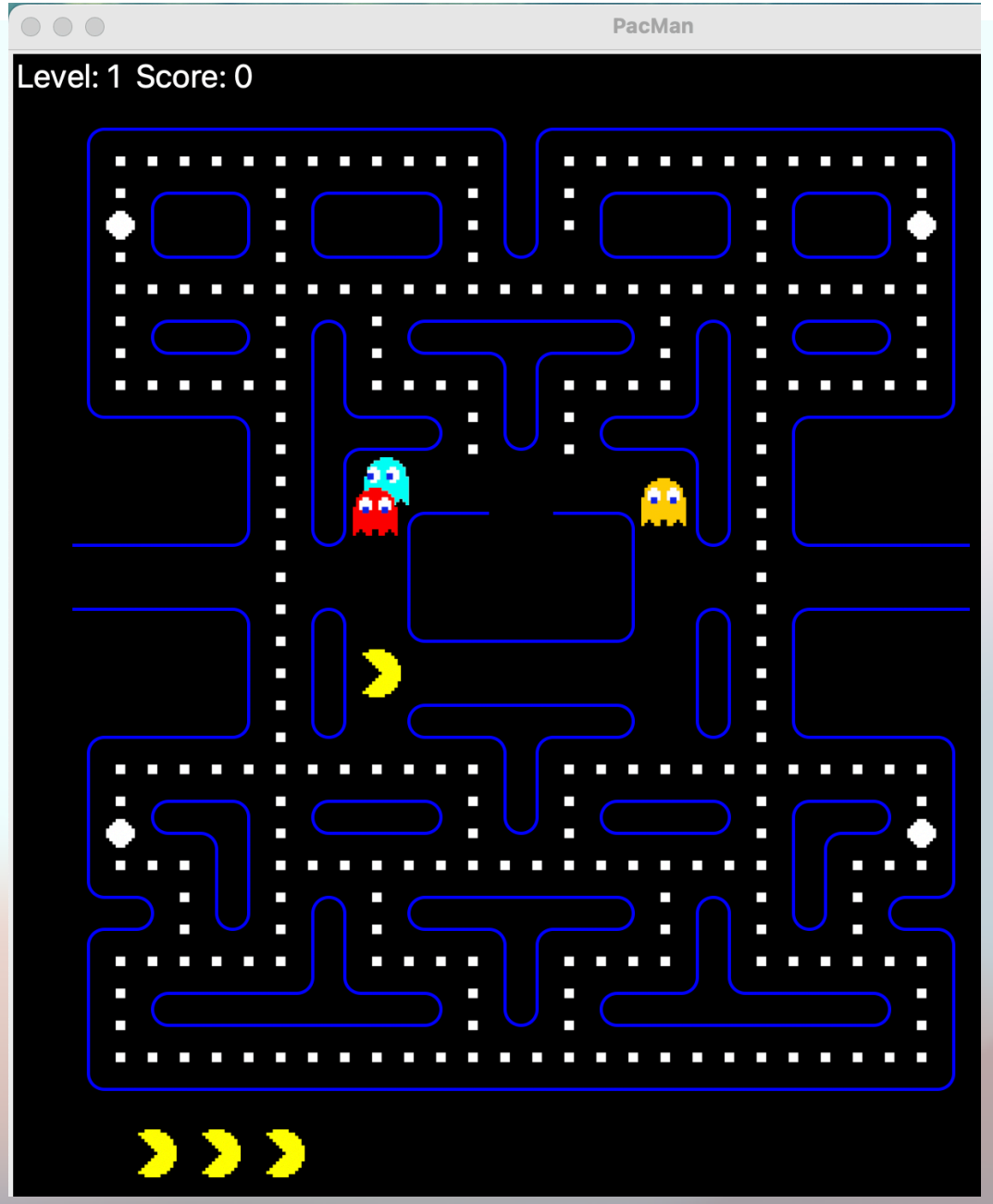
Assignments

Assignment 5: Write specification for Pacman game

1980 Maze video game

Special network version of the game

Task: write a better network protocol specification for this game



Other remarks

- For assessments 1 - 3 communicate and work with your peers
- If you have questions, it is likely your peers will have similar questions ➡️ ask via Moodle forum, not via e-mail
- Do not only use forum when you have questions; help your peers and answer their questions, too!
- However, do not ask or post solutions for summative (marked) assessments (4 & 5) ➡️ punishable offence
- Make use of forum and the drop-in clinics; do not suffer in silence!
- I know it is tempting to use Large Language Models (like ChatGPT, Gemini, Claude etc.), but there are risks:
  - Often you require programming experience to understand LLM outputs
  - More importantly, you require experience to identify wrong outputs!
  - It is not allowed to use LLMs for summative (marked) assessments!

Generative AI Use

- This module is classified as a Category 1 assessment according to the UCL Generative AI guidelines.
- Category 1: GenAI tools **cannot** be used.
- You must not use LLMs, generative AI tools, code assistants, or similar systems to complete (summative assessments of) this module. You are expected to work from the course material, and standard documentation only.
- Any use of GenAI tools for summative assessments in this module is considered academic misconduct and will be handled according to UCL's standard procedures. For further details, see the UCL GenAI assessment categories.

This Friday (October 9)

Pebble in the pond!



When? 11 am - 1 pm

Where?

Royal National Hotel

Next Tuesday (October 12)

Teamwork - Live Session

with Dr Charlotte Maughan Jones



When? 11 am - 1 pm

Where?

TBA

Now / Next hour ➡️ Programming foundations!